

SmartEdu Full-Stack School Management System

Internship Portfolio & Case Study
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Developed at Codec Technologies Pvt. Ltd.



Program Overview and Certification



Role: Full Stack Developer Intern



Duration: 2 Months (11/05/2026 – 11/07/2026)



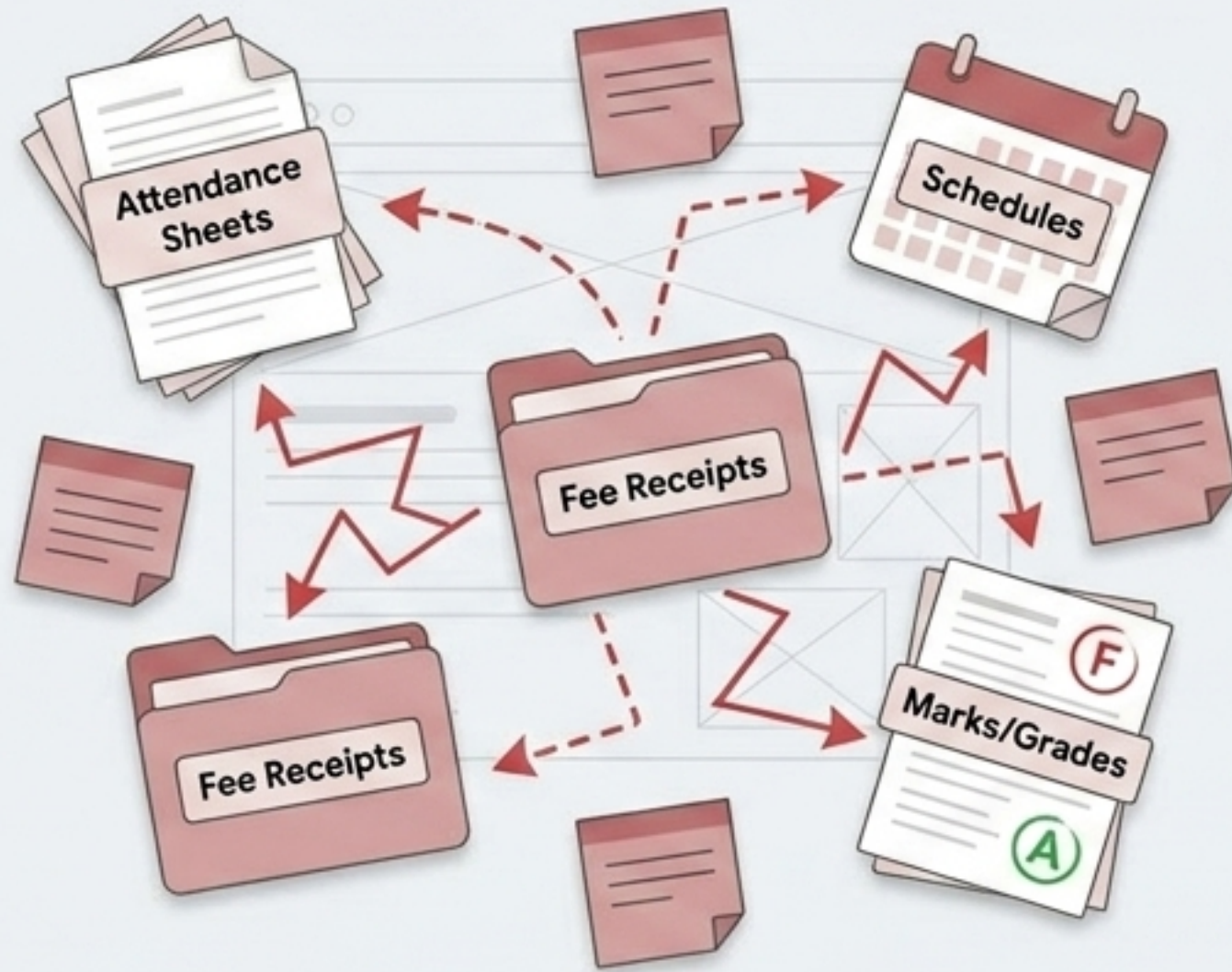
Program: AICTE & ICAC Approved Internship



Core Deliverable: Designed, developed, and deployed "SmartEdu", a secure, role-based School Management System integrating front-end interfaces with a PHP/MySQL backend.

The Educational Data Fragmentation Problem

Before



The Engineering Challenge

Data Silos: Information is difficult to maintain, retrieve, and update consistently.



The Engineering Challenge

Access Risks: Inability to securely restrict information based on the user's institutional role.



The Engineering Challenge

Operational Inefficiency: Disconnected systems prevent a unified view of student and school performance.

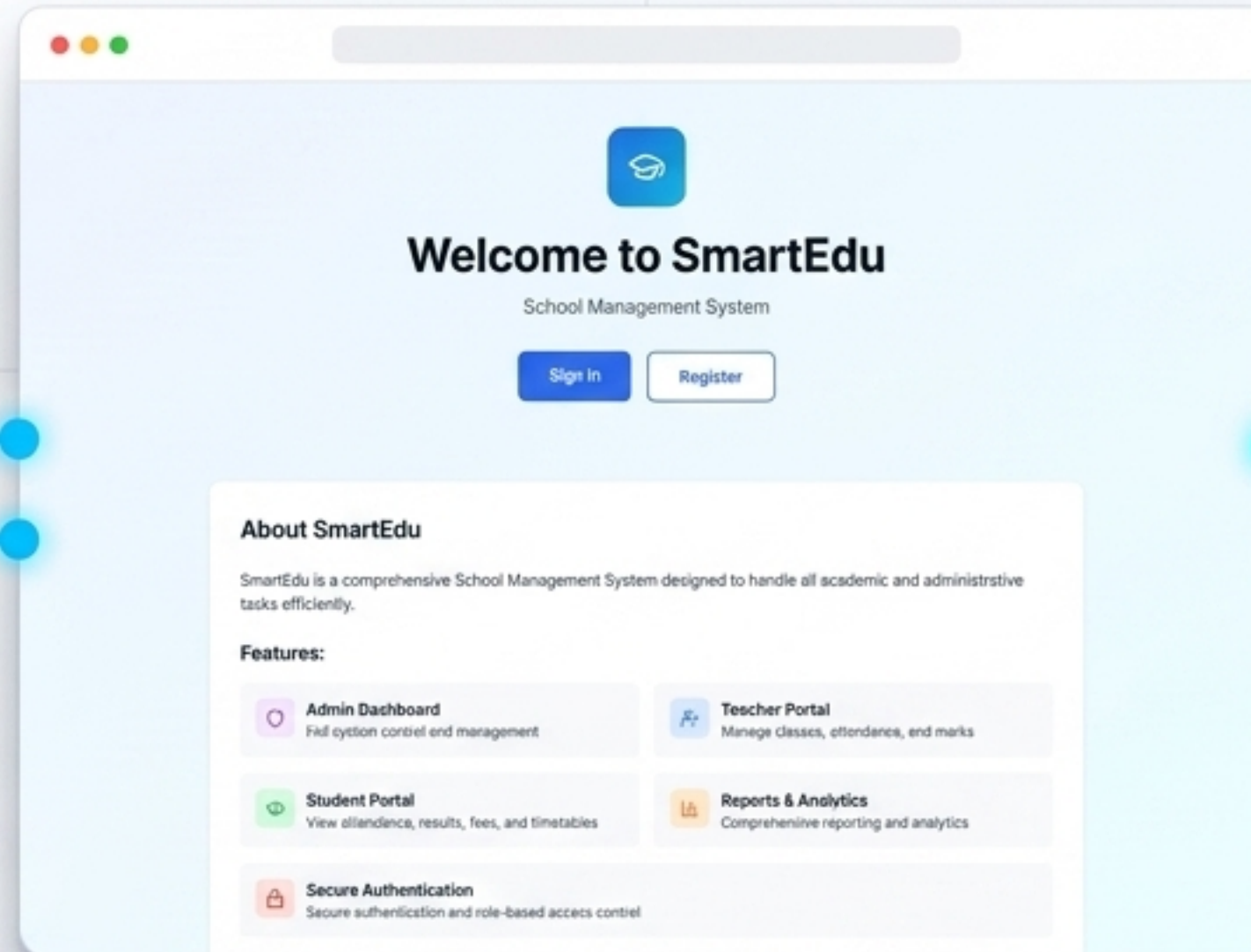
A Centralized Web-Based Environment



Centralized Database:
One definitive source for
all academic and
administrative data.



Unified Interface:
Consistent, responsive
design language across
all modules.



Role-Based Routing:
Secure gateways
directing users strictly to
authorized dashboards.

Role-Based Access Control Matrix

	 Administrator	 Teacher	 Student
System Configuration & Users	 Full Access	 Blocked	 Blocked
Fee Management & Records	 Manage	 Blocked	 View Own
Class Timetables	 Manage	 Upload/Manage	 View Own
Attendance Tracking	 View All	 Mark/Edit Class	 View Own
Examinations & Marks	 View All	 Enter/Grade	 View Own

Three-Tier Application Architecture

Development Environment (VS Code & GitHub)

Presentation Layer (Client)



Responsive UI, forms,
and dashboard layouts
rendered in the browser.

Application Layer (Server)



Core logic handling form
inputs, sessions,
authentication, RBAC, and
CRUD operations.

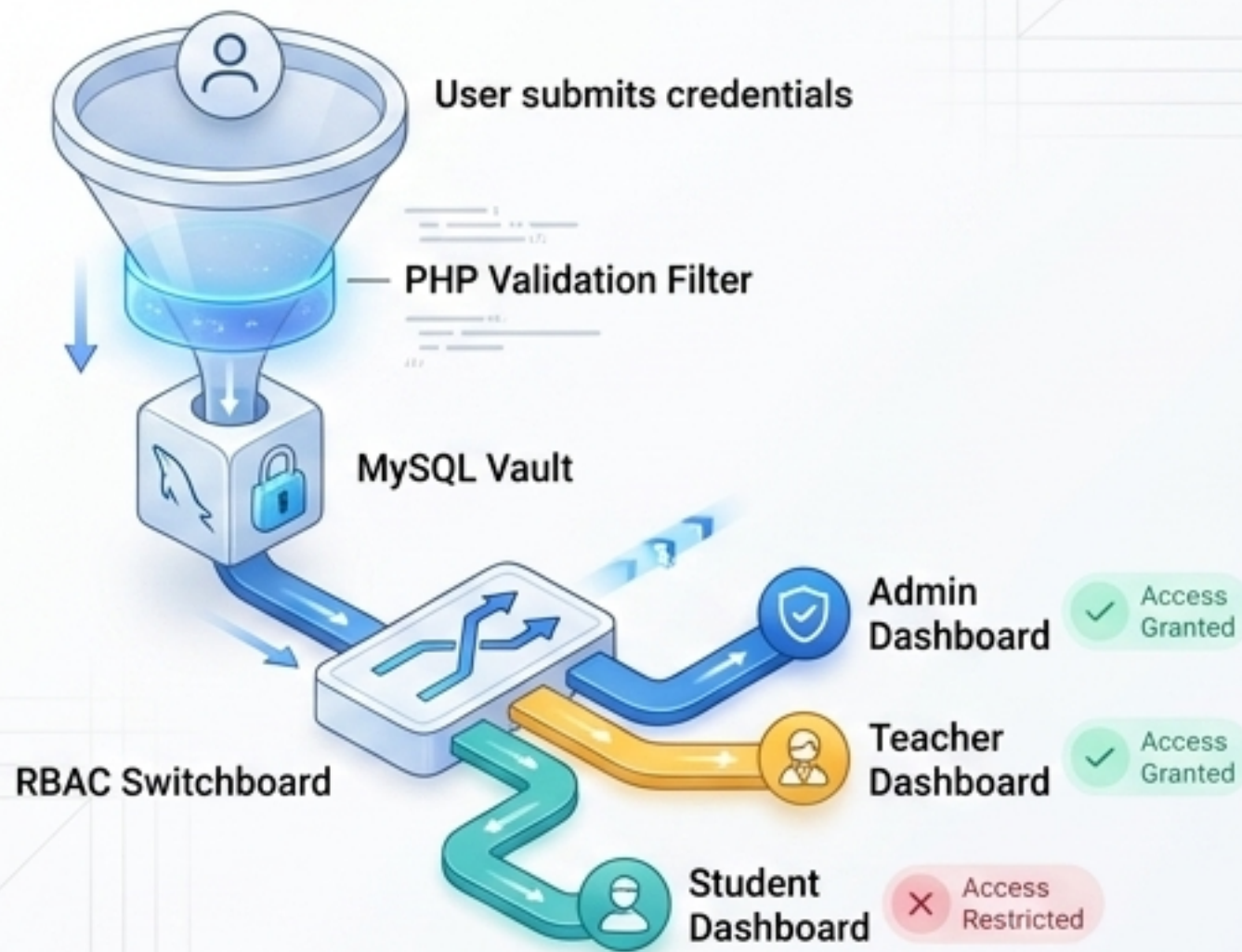
Data Layer (Database)



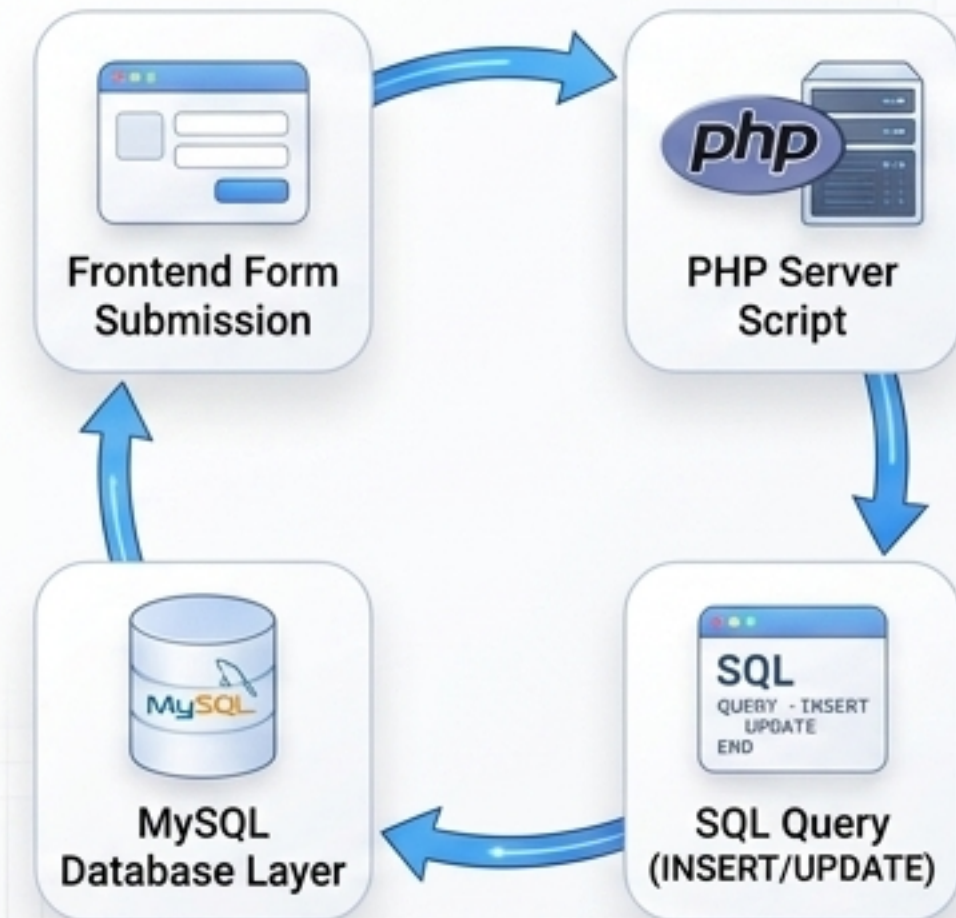
Persistent relational
storage, schema execution,
and structured SQL queries.

Engineering the Auth Funnel and CRUD Engine

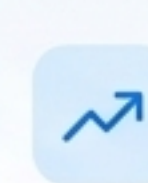
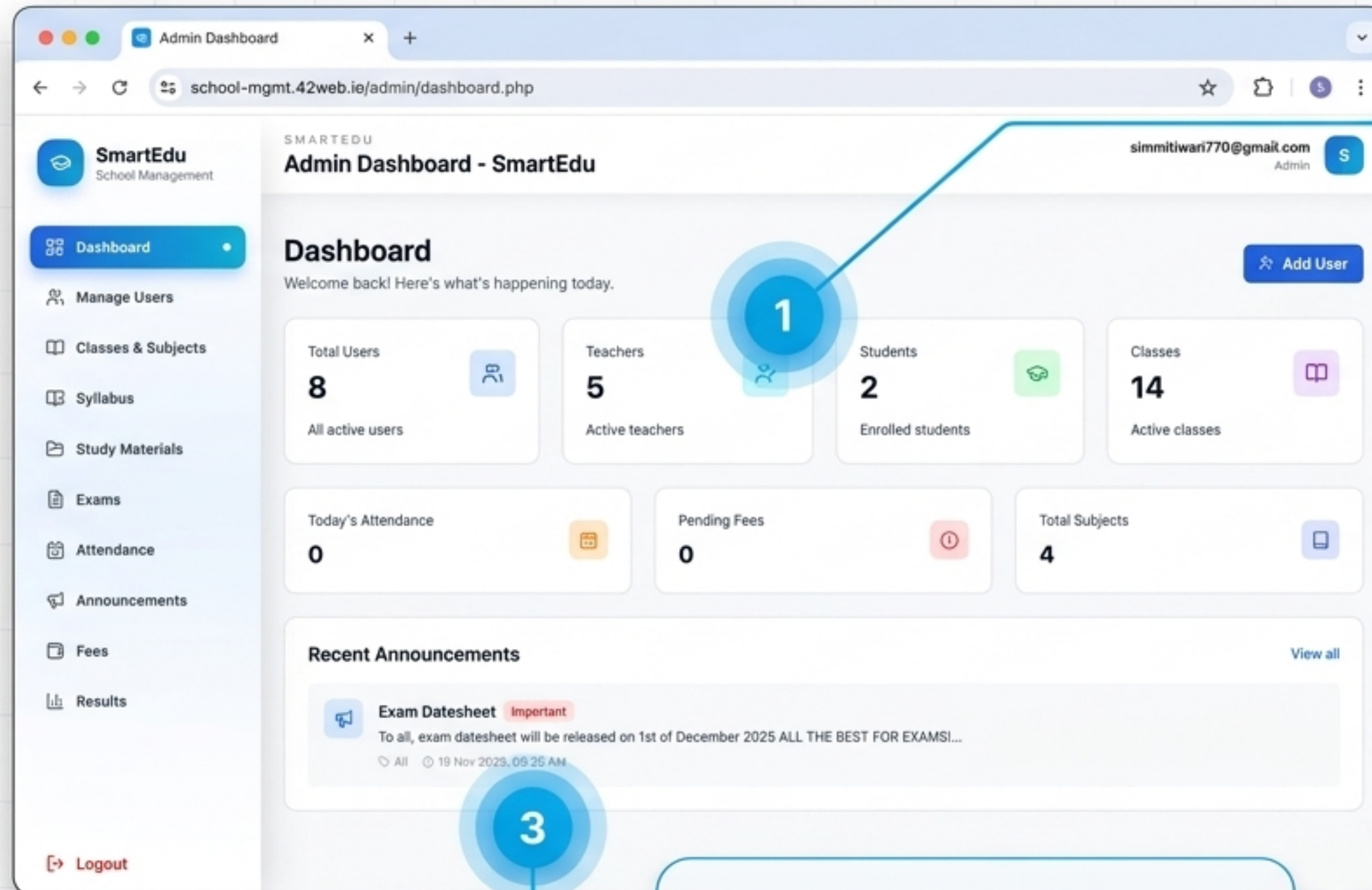
The Auth & RBAC Funnel



The CRUD Engine

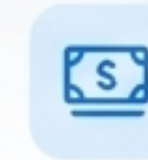
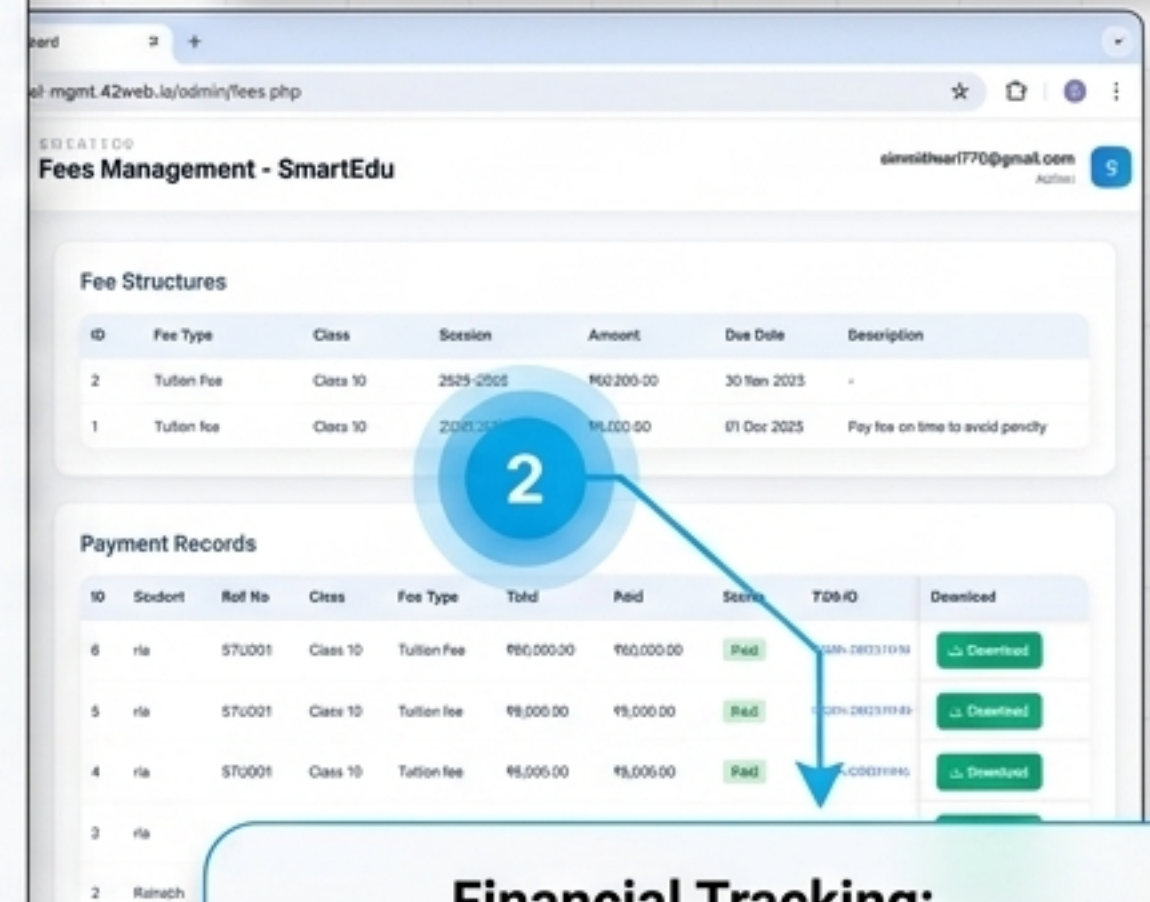


Administrative Control Center



Metrics Engine:

Real-time aggregation of active users, teachers, and students via SQL COUNT queries.



Financial Tracking:

Dedicated fee modules tracking tuition payments, generating dynamic receipts, and flagging pending balances.



Global Announcements:

System-wide broadcasting tools for datesheets and institutional alerts.

Teacher Academic Operations

The image displays two overlapping screenshots of the SmartEdu web application. The background screenshot shows the 'Teacher Dashboard - SmartEdu' for a user named Raji. It includes a sidebar with navigation options like Dashboard, Attendance, Marks Entry, Timetables, Syllabus, Study Materials, and Announcements. The main content area features a welcome message, summary cards for Classes (1) and Subjects (1), quick action buttons for Mark Attendance, Upload Timetable, Enter Marks, and Announcements, and sections for My Classes and Recent Announcements.

The foreground screenshot shows the 'Marks Management - SmartEdu' interface. It has a sidebar with Dashboard, Attendance, Marks Entry, Timetables, Syllabus, Study Materials, and Announcements. The main content area includes a 'Select Exam & Subject' section with a dropdown menu showing 'Final Exam - Class 10'. Below this is a 'Final Exam - Class 10' section with a table of student marks.

Three callouts highlight specific features:

- Contextual Filtering:** Dropdowns dynamically load specific class and subject rosters from the database. This is illustrated by the 'Select Exam' dropdown in the Marks Management interface, which shows 'Final Exam - Class 10'.
- Attendance State Management:** Intuitive toggle inputs for Present/Absent status updating relational tables in real-time. This is illustrated by the 'Mark Attendance' button in the Teacher Dashboard.
- Grade Book Engine:** Secure entry fields mapping numerical scores to specific student IDs and examination periods. This is illustrated by the 'Final Exam - Class 10' section in the Marks Management interface, which shows a table with student IDs (STUD01, STUD02) and names (ria, Rishabh).

Personalized Student Portal

1

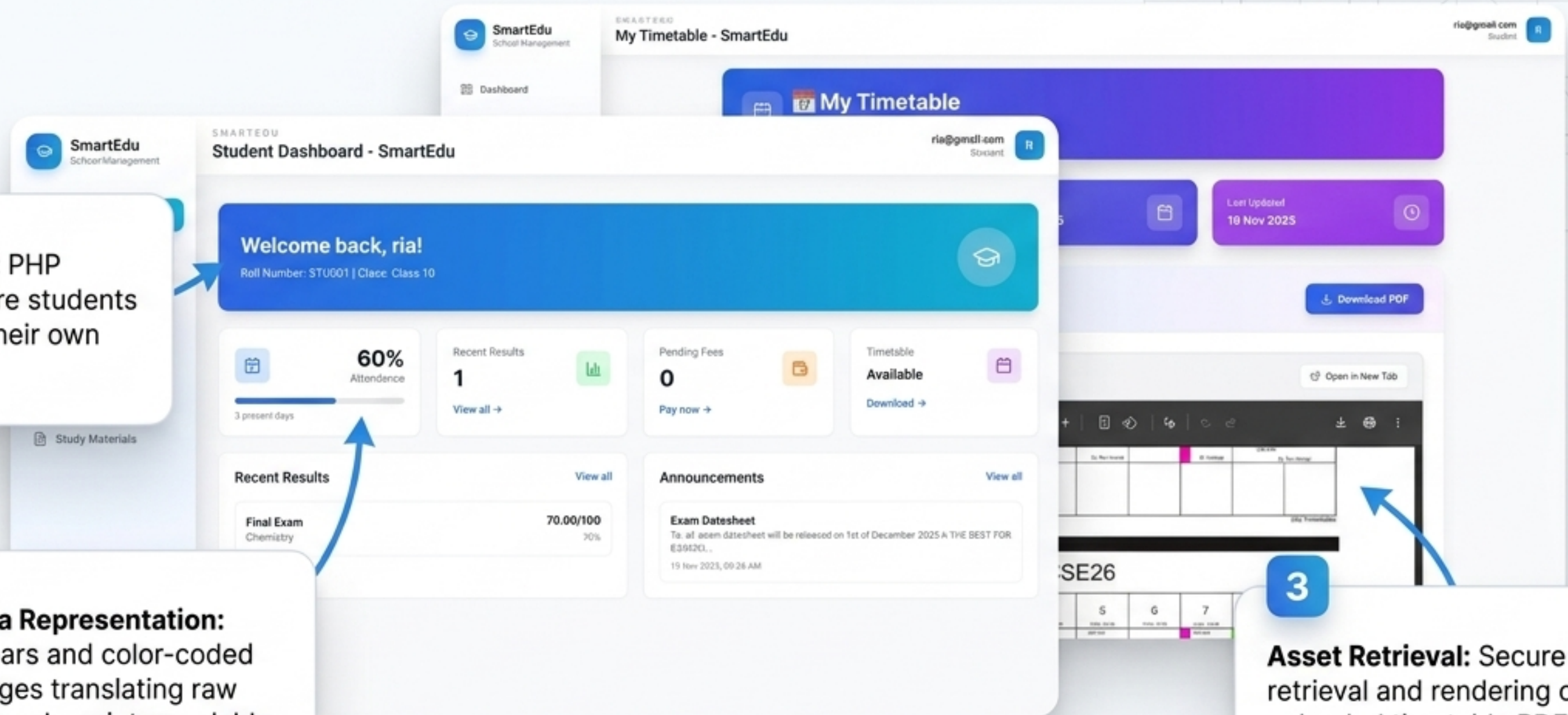
Session-Locked View: PHP session variables ensure students can strictly view only their own academic data.

2

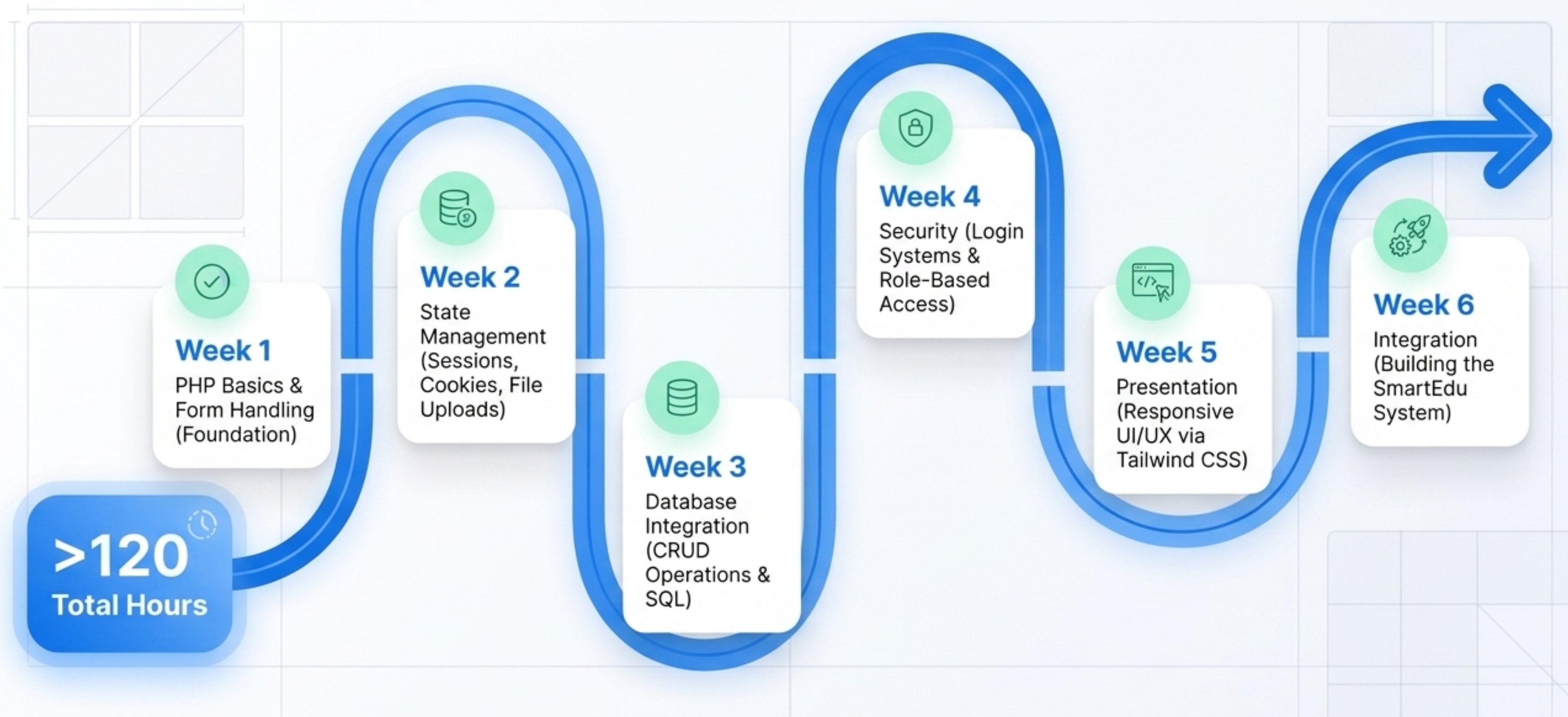
Visual Data Representation: Progress bars and color-coded status badges translating raw database numbers into readable attendance metrics.

3

Asset Retrieval: Secure retrieval and rendering of uploaded timetable PDF files directly within the browser tab.



The Six-Week Development Roadmap



Diagnostics and Problem Solving

! PROBLEMS

Roadblock 1: Complexities and edge-cases in backend Role-Based Access Control logic leading to improper dashboard routing.



Roadblock 2: Instability in database connections and failed execution of complex relational SQL queries.



✓ ANALYTICAL SOLUTIONS

Resolution 1: Consulted mentors and implemented systematic, step-by-step logic testing to isolate root causes and secure the routing paths.



Resolution 2: Refined the overall MySQL schema for better relational integrity and integrated robust PHP error-handling to ensure stable connectivity.



Synthesis: From Theory to Ecosystem



The internship bridged the gap between isolated academic programming exercises and the reality of building interconnected, role-aware software architectures, supported by collaborative debugging and version control (GitHub).